



# Montassar Laboudi

## AI & Signal Processing Engineer

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Saint-Paul-lès-Durance (13115), France

### PROFILE

**AI and signal processing engineer**, graduated from **Bretagne INP – ENIB** (Signal Processing & Telecommunications) and **ENIT** (Telecommunications Engineering, specialization: Data Science for Embedded Communications).

Hands-on experience designing and deploying ML/DL systems, processing real-world experimental data, and building production-ready AI applications — from research environments to cloud infrastructure.

### SKILLS

**AI & Deep Learning:** Supervised/unsupervised learning, CNN, LSTM, Transformers, transfer learning, model training & evaluation, hyperparameter optimization, anomaly detection, computer vision, NLP.

**Generative AI & LLMs:** RAG pipelines, LLM integration, prompt engineering, vector databases, embeddings, OpenAI API, LangChain, multimodal models.

**Signal & Image Processing:** FFT, STFT, wavelet transforms, spectral analysis, audio processing, Point cloud, segmentation, feature extraction, infrared imaging, OpenCV.

**MLOps & Development:** PyTorch, TensorFlow/Keras, scikit-learn, Docker, Git, Linux, Python, MATLAB, SQL, C/C++, TypeScript.

### EXPERIENCE

#### CEA — French Alternative Energies and Atomic Energy Commission

Mar. 2025 – Aug. 2025

Engineering Intern — Experimental Data Analysis & Scientific Software

Cadarache, France

- Designed and developed a scientific software application for processing, visualization, and quantitative analysis of 3D experimental data from confocal microscopy within the WEST nuclear fusion project; built automated pipelines for preprocessing, segmentation, and feature extraction on large, noisy real-world datasets.

– **Stack:** Python, PyQt, NumPy, SciPy, 3D image processing, segmentation, scientific computing.

#### dB.Sense

Jun. 2024 – Aug. 2024

Engineering Intern — Embedded AI & Audio Signal Processing

Tunis, Tunisia

- Built a complete real-time sound event detection system — from audio acquisition and mel-spectrogram feature extraction to deep learning inference and edge deployment on Raspberry Pi under strict compute, memory, and latency constraints.

– **Stack:** Python, TensorFlow, DSP, mel-spectrograms, Raspberry Pi, embedded Linux, Docker, real-time inference.

#### Sharing Technologies

Jun. 2023 – Aug. 2023

Software Development Intern

Tunis, Tunisia

- Built a real-time equipment monitoring application with live status tracking via SignalR in a C#/.NET client-server architecture.

### PROJECTS

#### AnimeGPT – Multimodal RAG Chatbot [[animegpt-rag-chatbot.vercel.app](#)]

2026

Personal project — Generative AI · RAG · Multimodal · Production deployment

- End-to-end RAG system with GPT-4o multimodal inference: automated data scraping, embedding generation, vector indexing, semantic retrieval, token-by-token streamed generation, and vision-based image identification — deployed to production on Vercel.
- **Stack:** Next.js, TypeScript, OpenAI GPT-4o, LangChain.js, Astra DB (vector DB), Vercel AI SDK, Puppeteer.

#### Deep Learning Image Classification – ENIB

2025

Academic project — Computer Vision · Deep Learning · Model Evaluation

- Trained and benchmarked convolutional neural networks for multi-class image classification across multiple datasets; implemented data augmentation, learning rate scheduling, and hyperparameter tuning with reproducible experiments and state-of-the-art comparison.
- **Stack:** Python, PyTorch, CNN, transfer learning, data augmentation, cross-validation, classification metrics.

#### Acoustic Signal Analysis – CEA / HADES (WEST Project)

2025

R&D project — DSP · Experimental Signal Processing · Scientific Analysis

- Processed and analyzed noisy acoustic signals from plasma-facing components of the WEST fusion device; applied FFT, STFT, and wavelet transforms to extract physical signatures and characterize thermal and hydraulic phenomena from real experimental measurements.

– **Stack:** Python, MATLAB, FFT, STFT, wavelet analysis, filtering, correlation, spectral feature extraction.

### EDUCATION & CERTIFICATIONS

#### Bretagne INP – École Nationale d'Ingénieurs de Brest (ENIB)

2024 – 2025

Master's Degree – Signal Processing and Telecommunications

Brest, France

#### École Nationale d'Ingénieurs de Tunis (ENIT)

2022 – 2025

Telecommunications Engineering Degree – Data Science for Embedded Communications

Tunis, Tunisia

#### Certifications:

OCI Generative AI Professional · OCI AI Foundations · NVIDIA Deep Learning · Data Analytics Professional